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EDITORIAL

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GNE myopathy: Don't sleep on the platelets

GNE myopathy (formerly known as Nonaka myopathy, distal myopathy with rimmed vacuoles, and hereditary inclusion body myopathy) is an early-adult onset distal myopathy with preferential involvement of the anterior compartment of the leg and myopathological evidence of rimmed vacuoles.[1] Rarely, *GNE* myopathy patients may present with limb-girdle weakness.[2] Quadriceps are selectively spared in both the distal and proximal phenotypes, even in the advanced stage of the disease. Respiratory weakness is uncommon and, when present, occurs late and with mild severity. The disease is inherited in an autosomal recessive fashion and found worldwide, however, is relatively more common in Japan[3] and the Persian Jewish community.[4] *GNE* encodes uridine diphospho-N-acetylglucosamine (UDP-GlcNAc) 2-epimerase/N-acetyl-mannosamine (ManNAc) kinase (GNE/MNK), a bifunctional enzyme responsible for the rate-limiting steps in sialic acid synthesis from glucose.[5] Myopathy in patients with mutations in *GNE* is thought to result from hyposialylation of muscle glycans due to impaired or reduced GNE/MNK activity.[1]

Recently, the phenotypic spectrum of *GNE*-related disorders has expanded to include thrombocytopenia identified in isolation[6-8] or in conjunction with typical myopathic features.[3,9-11] The former was demonstrated in patients predominantly within the first decade of life, the majority in the neonatal period, prior to the expected age of onset in *GNE* myopathy. Therefore, whether the thrombocytopenia observed in those cases is truly isolated remains unresolved. Additionally, whether subclinical features of myopathy exist in those patients remains largely unknown. Among patients with *GNE* mutations and thrombocytopenia, bleeding complications are typically mild and non-life threatening, frequently manifesting as petechiae or epistaxis, despite significantly low platelet counts in some.[8] Only rarely are platelet transfusions required in the setting of severe bleeding.

Why does thrombocytopenia occur in patients with *GNE* mutations? Sialic acid normally caps platelet surface glycoproteins and is key to several cell–cell interactions, including platelet aggregation and adhesion.[12] Desialylation exposes platelet surface glycans, which then serve as ligands for lectin receptors on macrophages and hepatocytes, which clear the desialylated platelets from circulation, leading to reduced platelet counts.[13] This is in contrast to the autoantibody-mediated clearance of platelets via splenic macrophages in immune thrombocytopenia (ITP). Mutations observed in *GNE*-thrombocytopenia patients with or without myopathy clustered in the MNK domain, spanning a segment of approximately 130 amino acids from p. Asp444 through p.Gly578.[8] Patients were generally homozygous for mutations in this region; however, a minority were compound heterozygous with at least one mutation located in the hotspot or just downstream of it. Mutations in this region likely impact adenosine diphosphate (ADP) or substrate binding, but may also disrupt normal protein folding. Surface sialylation occurs across hematopoietic cell lines; thus, it still remains unclear as to why platelets appear particularly vulnerable to reduced MNK activity or altered substrate binding. Not all *GNE* myopathy patients have undergone platelet evaluation; therefore, whether this 130-amino acid segment in the MNK domain is truly a hotspot for thrombocytopenia remains to be determined.

In this issue of *Muscle and Nerve*, Mori-Yoshimura and colleagues report their prospective, observational experience following platelet indices, measures of strength, sleep-disordered breathing, and cardiac function in 33 Japanese patients with genetically confirmed *GNE* myopathy.[14] Among 28 patients who underwent platelet evaluation, 3 had thrombocytopenia at baseline, with 2 carrying prior diagnosis of ITP in childhood without significant bleeding or requirement for ongoing medical therapy. The prevalence of presumed ITP (10.7%) in this cohort is much greater than expected for the Japanese population (<0.02%) and, given our current understanding, likely reflects misattribution of ITP to an inherited etiology for thrombocytopenia due to mutations in *GNE*. Awareness of thrombocytopenia occurrence in *GNE* myopathy is of importance, so as to avoid the use of potentially harmful therapies (i.e., chronic prednisone) or invasive procedures (i.e., splenectomy) that have been shown to be ineffective in restoring platelet count in *GNE*-thrombocytopenia.[8,9,11]

More notable in the reported cohort is the high frequency (17/26) of platelet-associated immunoglobulin G (PA-IgG), even among those without overt thrombocytopenia. PA-IgG is a non-specific marker of shortened platelet lifespan, which may be found in both immune-mediated and non-immune mediated thrombocytopenic disorders.[15] Evaluation for PA-IgG or other platelet-specific autoantibodies in suspected ITP is not guideline-recommended due to low sensitivity and specificity and ITP remains a diagnosis of exclusion[16]; however, Mori-Yoshimura et al highlight the potential utility of PA-IgG in patients with *GNE* myopathy. Intriguingly, PA-IgG positivity was associated with significantly lower baseline summed manual muscle testing (MMT) score, grip power, and platelet counts in their cohort, suggesting its potential as a biomarker of disease severity. Furthermore, PA-IgG titers may be more sensitive than platelet counts for the increased platelet turnover in patients with *GNE* mutations

Abbreviations: ADP, adenosine diphosphate; ITP, immune thrombocytopenia; ManNAc2-epimerase/N-acetyl-mannosamine; MMT, manual muscle testing; MNK, N-acetyl-mannosamine kinase; PA-IgG, platelet-associated immunoglobulin G; UDP, uridine diphospho-N-acetylglucosamine.

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without overt thrombocytopenia. It appears logical to assume, given the fairly stereotyped progression in *GNE* myopathy, that PA-IgG positivity should also associate with earlier age of onset or longer disease duration; however, this was not borne out statistically possibly due to the small cohort size, although a trend toward significance ($P = .054$) was noted with the former.

The exact pathogenesis of increased PA-IgG in *GNE* myopathy is uncertain; however, desialylation of platelet membrane glycoproteins may provide an antigenic target for IgG.[17] Whether PA-IgG itself then contributes to platelet clearance, in addition to the aforementioned lectin-receptor mediated clearance, is unknown. It likewise remains to be seen whether PA-IgG may similarly act as a biomarker of disease severity in other inherited myopathies associated with thrombocytopenia, such as *STIM1*-tubular aggregate myopathy, wherein the mechanism of thrombocytopenia is unrelated to sialylation but rather due to abnormal calcium ion handling.[18] Regardless, PA-IgG remains a potential biomarker of interest in future prospective studies and treatment trials in *GNE* myopathy.

Not to be overlooked is the high frequency of sleep apnea in the cohort (14/19), predominantly obstructive, which occurred independently of respiratory dysfunction, MMT scores, age, or body mass index and in the absence of dysphagia or aspiration pneumonia.[14] This highlights the importance of early surveillance for sleep-disordered breathing in all patients with *GNE* myopathy at the time of initial diagnosis. Consistent with prior characterization of the disease[1] was the lack of cardiomyopathy and arrhythmia or significant cardiac complication over the 5-year study period, reiterating the notable lack of cardiac involvement in *GNE* myopathy.

There are limitations to the current study. The sample size is small, and only 22 patients completed the full 5 years of follow-up, some of whom missed scheduled evaluations. Of those 12 patients who did not complete the full period of follow-up, 7 were ambulant, potentially skewing the results of the study toward those with more severe disease. Only 26 patients underwent PA-IgG evaluation, limiting the assessment of its association with disease duration and age of onset. Sleep studies were performed in only 19 patients, where selection bias may have contributed to more severely affected patients completing the evaluation and, thus, limiting our understanding of the spectrum of sleep disordered breathing in those earlier in their disease course.

Mori-Yoshimura and colleagues highlight that *GNE* myopathy can now assuredly take its place among the expanding list of inherited myopathies associated with hematological abnormalities.[19] Further prospective evaluation of the significance of PA-IgG in larger patient cohorts is needed, but PA-IgG appears to be a promising biomarker in this disorder.

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CONFLICTS OF INTEREST

None of the authors have any conflict of interest to disclose.

AUTHOR CONTRIBUTIONS

Grayson Beecher: concept and design, drafting. **Teerin Liewluck:** concept and design, and critical revisions of the manuscript.

ETHICAL PUBLICATION STATEMENT

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

DATA AVAILABILITY STATEMENT

This is an editorial article. No data generated from this manuscript.

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REFERENCES

1. Carrillo N, Malicdan MC, Huizing M. GNE myopathy: etiology, diagnosis, and therapeutic challenges. *Neurotherapeutics*. 2018;15(4):900-914.
2. Park YE, Kim HS, Choi ES, et al. Limb-girdle phenotype is frequent in patients with myopathy associated with GNE mutations. *J Neurol Sci*. 2012;321(1–2):77-81.
3. Mori-Yoshimura M, Hayashi YK, Yonemoto N, et al. Nationwide patient registry for GNE myopathy in Japan. *Orphanet J Rare Dis*. 2014;9:150.
4. Argov Z, Eisenberg I, Grabov-Nardini G, et al. Hereditary inclusion body myopathy: the middle eastern genetic cluster. *Neurology*. 2003; 60(9):1519-1523.
5. Eisenberg I, Avidan N, Potikha T, et al. The UDP-N-acetylglucosamine 2-epimerase/N-acetylmannosamine kinase gene is mutated in recessive hereditary inclusion body myopathy. *Nat Genet*. 2001;29(1):83-87.
6. Revel-Vilk S, Shai E, Turro E, et al. GNE variants causing autosomal recessive macrothrombocytopenia without associated muscle wasting. *Blood*. 2018;132(17):1851-1854.
7. Li X, Li Y, Lei M, et al. Congenital thrombocytopenia associated with GNE mutations in twin sisters: a case report and literature review. *BMC Med Genet*. 2020;21(1):224.
8. Bottega R, Marzollo A, Marinoni M, et al. GNE-related thrombocytopenia: evidence for a mutational hotspot in the ADP/substrate domain of the GNE bifunctional enzyme. *Haematologica*. 2021. doi: 10.3324/haematol.2021.279689
9. Izumi R, Niihori T, Suzuki N, et al. GNE myopathy associated with congenital thrombocytopenia: a report of two siblings. *Neuromuscul Disord*. 2014;24(12):1068-1072.

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10. Zhen C, Guo F, Fang X, Liu Y, Wang X. A family with distal myopathy with rimmed vacuoles associated with thrombocytopenia. *Neurol Sci*. 2014;35(9):1479-1481.
11. Paul P, Liewluck T. Distal myopathy and thrombocytopenia due to a novel GNE mutation. *J Neurol Sci*. 2020;415:116954.
12. Sørensen AL, Rumjantseva V, Nayeb-Hashemi S, et al. Role of sialic acid for platelet life span: exposure of beta-galactose results in the rapid clearance of platelets from the circulation by asialoglycoprotein receptor-expressing liver macrophages and hepatocytes. *Blood*. 2009; 114(8):1645-1654.
13. Grozovsky R, Hoffmeister KM, Falet H. Novel clearance mechanisms of platelets. *Curr Opin Hematol*. 2010;17(6):585-589.
14. Mori-Yoshimura M, Kimura A, Tsuru A, et al. Assessment of thrombocytopenia, sleep apnea, and cardiac involvement in GNE myopathy patients. *Muscle Nerve*. 2022;65(3):284-290.
15. Mueller-Eckhardt C, Kayser W, Mersch-Baumert K, et al. The clinical significance of platelet-associated IgG: a study on 298 patients with various disorders. *Br J Haematol*. 1980;46(1):123-131.
16. Neunert C, Lim W, Crowther M, Cohen A, Solberg L Jr, Crowther MA. The American Society of Hematology 2011 evidence-based practice guideline for immune thrombocytopenia. *Blood*. 2011;117(16):4190-4207.
17. Kotzé HF, van Wyk V, Badenhorst PN, Heyns AD, Roodt JP, Lötter MG. Influence of platelet membrane sialic acid and platelet-associated IgG on ageing and sequestration of blood platelets in baboons. *Thromb Haemost*. 1993;70(4):676-680.
18. Lacruz RS, Feske S. Diseases caused by mutations in ORAI1 and STIM1. *Ann N Y Acad Sci*. 2015;1356:45-79.
19. Beecher G, Fleming MD, Liewluck T. Hereditary myopathies associated with hematological abnormalities. *Muscle Nerve*. 2021. doi:10.1002/mus.27474

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